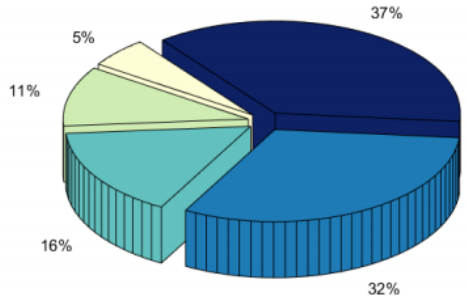
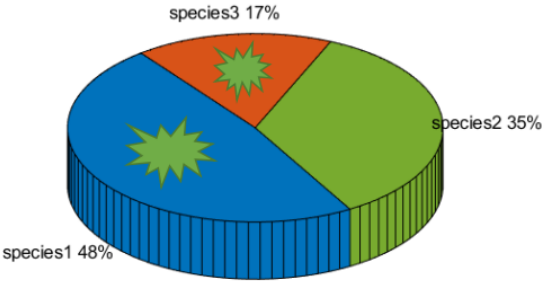
美赛速成系列|美赛O奖论文复现

——美赛中数据图绘制

> 之前说过，美赛论文美观度以及图片绘制是拿奖的关键，这篇笔记带大家来复现**美赛O奖优秀论文的数据图的绘图**，并且整理了**Matlab绘图代码**，有运行不出来也可以问学姐哦~

饼状图






```
%% 绘制饼图
load('color cell1.mat')
mycolor1=color_cell1{1,1};
% data=0.1607843137254900.6156862745098040.57254901960data=[1
figure
pie3(data,ones(1,length(data)))
colormap(mycolor1)
set(gca，'looseInset'[0000);%去除图片的白边
%%
figure
labels=["species1 48%""species2 35%","species3 17%"];
pie3([48,35,17l,labels)
0.0039color1 set=f
0.4471
0.7373
```

```
0.4706
0.6706
0.1882
0.84710.3255set(gca,'looseInset', [0000]);%去除图片的白边
colormap(color1 set)
```

折线图

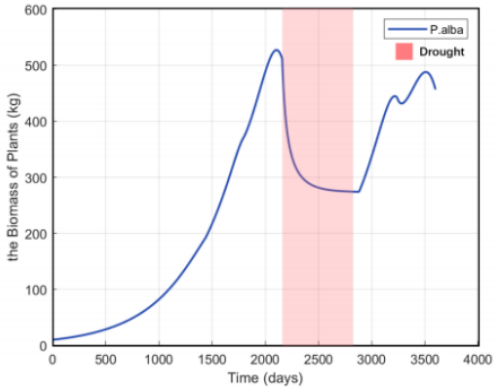


Figure 5: Biomass of *Pinus massoniana Lab*

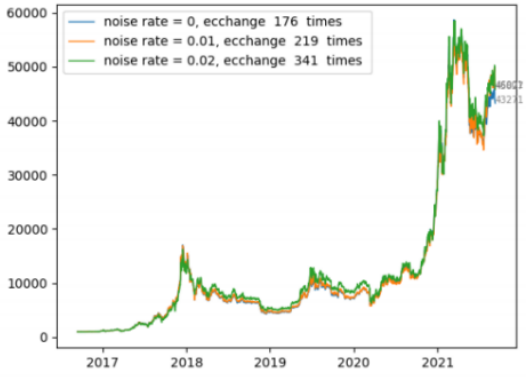
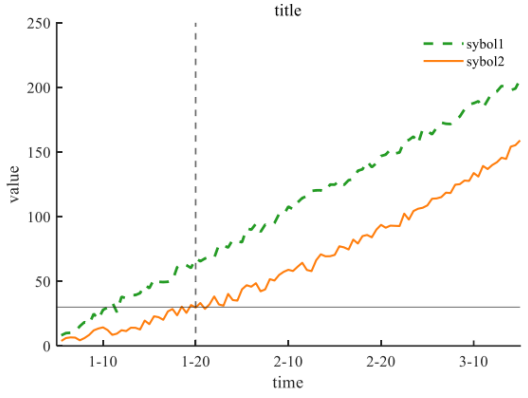
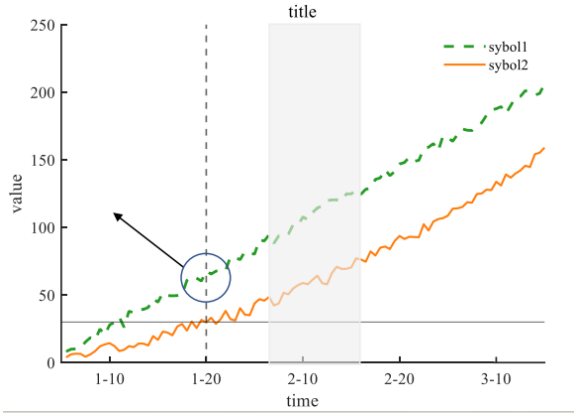


Figure 12: Results on price data with Gaussian noise

> 首先我们用**matlab代码生成折线图**，并且折线图的配色可以选择与O奖论文中一致的配色，比较美观好看，然后在**PPT中可以进一步处理绘图**，比如添加加箭头、矩形等操作突出重点，得到下面右图的结果。

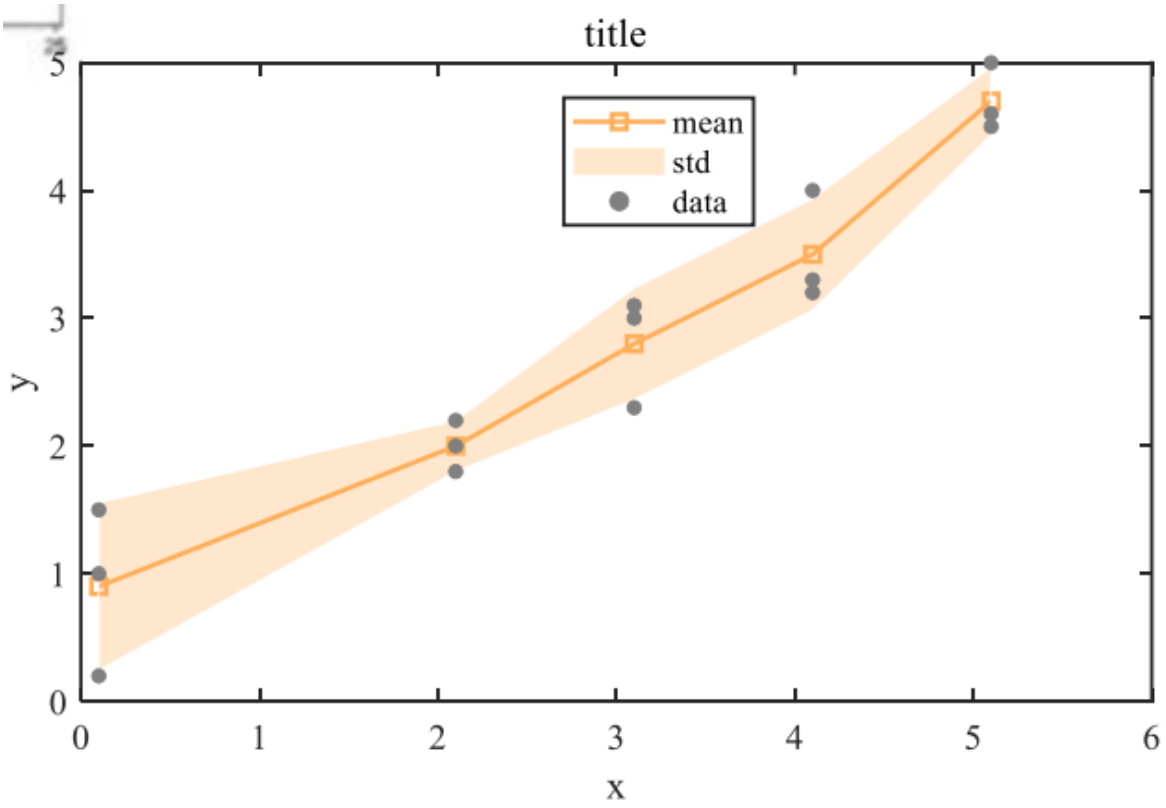




```
% 绘图案例
%% 折线图
%% 图框的基本设置
clc;clear;close all;
load('color list.mat')
load('折线图测试数据.mat')
figure('Position',[100,100,600,400])%整个框大小设置
x1=[1:100];
y1=2*x1+10*rand(1,length(x1));y2=0.01*x1.^2+0.5*x1+10*rand(1,
color_index=[1,3,4,7,8,9]; %颜色选取
fu={'_*','_^','-0',’-d','-p'};%符号
% %单根折线图
% plot(x1,y1,'--','Linewidth',2,'color',color list(color_inde
%保持在一个图窗内
% hold on
% plot(x1,y2,'--','LineWidth',2,'color',color list(color inde
% hold on
color_set=[0.145  0.6118  0.1451;1.0000  0.4980  0.0
% 单根折线图
"--', 'LineWidth'2,'Color'color set(1:))plot(x1,y1,
%保持在一个图窗内
hold on
plot(x1,y2,'-','Linewidth',1.5,'Color',color_set(2,:))hold on
xlabel('time')
ylabel('value')
title('title')
xticks([10:20:100]);
% xticks([10,30,50,70,90]);
xticklabels({'1-10','1-20','2-10','2-20','3-10'})
```

> 代码太长，这里仅展示部分

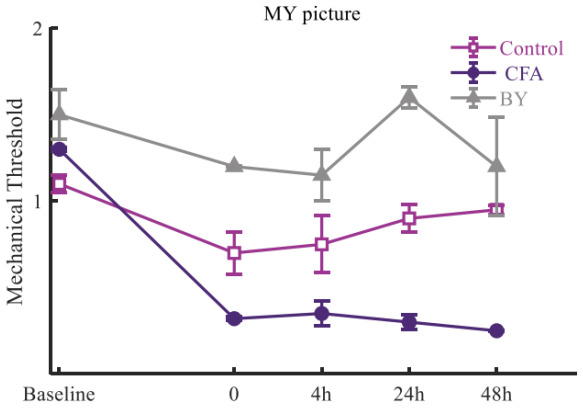
单个带误差棒的折线图

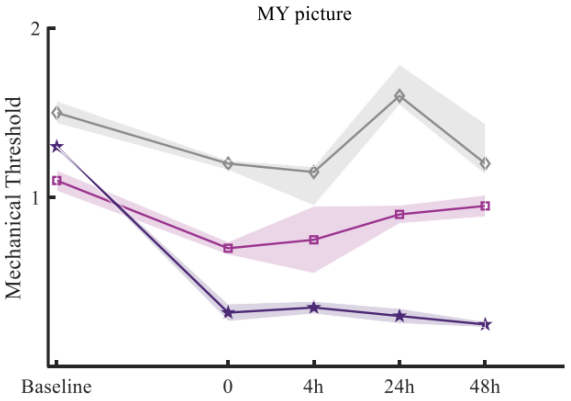



```
%%单个误差带的折线图
data_test=[1,2,3,4,5;1.5,1.8,2.3,3.2,4.5;0.2,2.2,3.1,3.3, 4.6
x=[0.1,2.1,3.1,4.1,5.1];%x轴数据
y1=dmean; %y轴数据
%数据上限low1=dmean-dstd;
%数据下限high1=dmean+dstd;
figure('Position'[200,200，600，3501)
0.3490];color_set=[1.00000.6863
plot(x,y1,'-s','Color',color_set(1,:),'LineWidth',1.5)hold on
h1.Facecolor=color_set(1,:);%定义区间的填充颜色
h1.EdgeColor =[1,1,1];%边界颜色设置为白色
%设置透明色alpha(0.3)
scatter(x,data test,20,'o','filled','MarkerFaceColor',[0.5,0.
set(gca,'FontName','Times New Roman',"FontSize",12,"Linewidth
% box off
legend('mean','std','data')
xlabel('x'
ylabel('y')
title('title')
```

多个带误差棒/误差带的折线图

> 1.图中包含均值和标准差，适用于美赛，表示的信息更多
>
> 2.如果有多次实验的绘图可以考虑这种带误差棒的折线图






```
%% 带误差棒的折线图
clc;clear;close all;load('color list.mat')color=0.61176470588
% coor=color list(index，:);
% data test=[1,2,3;1.1,1.9,2.9;1.2,2.1,3.1];
% dmean=mean(data test);
% dstd=std(data test);
号
% x=[0.1,2.1,3.1];%x轴数据
号
%y1=dmean; %y轴数据
。。
%数据上限% low1=dstd;
% high1=dstd;%数据下限
号
% errorbar(x,y1,low1,high1,'-s','color',color(1,:),'MarkerSiz
% hold on
x=[0.1,2.1,3.1,4.1,5.11;%x轴数据
%y轴数据y1=[1.1,0.7,0.75,0.9,0.95];
low1=0.2*rand(1,length(y1));%数据上限high1=low1;%数据下限
y2=[1.3,0.32,0.35,0.3,0.251;low2=0.1*rand(1,length(y2))high2=
y3=[1.5,1.2,1.15,1.6,1.2];
low3=0.3*rand(1,length(y3));
```

```
high3=low3 ;
% 画图
%调整整个图的范围
figure('units''pixels'
'Position'，「1001006003757);%'Color'整个连接线的颜色，'MarkerSiz
errorbar(x,y1,low1,high1,'-s','Color',color(1,:),'MarkerSizeL
errorbar(x,y2,low2,high2,'-o','Color',color(2,:),'MarkerSize
'LineWidth'，2'capSize'10)
hold on
errorbar(x,y3,low3,high3,'-','Color',color(3,:),'Markersize''
ylabel('Mechanical Threshold')
```